Scientific method	Observe	Hypothesize			Validate
ML Component	Train data	Model	Loss	Algorithm	Test Data
	$(x,y) \in \mathbb{R} \times \mathbb{R}$	$\hat{y} = \hat{w}x$	$l = y - \hat{y}$	Fixed-step ($\hat{w} \leftarrow \hat{w} \pm \alpha$)	
Robbins-Monro	$(x_1,y_1) \dots (x_n,y_n)$			Robbins-Monro ($\hat{w} \leftarrow \hat{w} + \alpha l$)	
			$l = \frac{1}{n} \sum_{i \le n} (y_i - \hat{y}_i)$		
Modern ML	many features $x_i = (x_{i1}, \dots, x_{id})$	$\hat{y}_i = \hat{w}^T x_i$	Zoo of loss functions	Gradient descent $\hat{w} \leftarrow \hat{w} - \alpha \nabla l$	
		Neural networks			